

Math course, which the student has taken.

b) Someone has visited every country in the world except Libya.

$V(x, y)$: Person x has visited country y .

Domain of x : Set of all persons of the world

Domain of y : Set of all countries of the world

$$\exists x (\neg V(x, \text{Libya}) \wedge \forall y ((y \neq \text{Libya}) \Rightarrow V(x, y)))$$

$$\neg \exists x (\neg V(x, \text{Libya}) \wedge \forall y ((y \neq \text{Libya}) \Rightarrow V(x, y)))$$

$$\equiv \forall x \neg (\neg V(x, \text{Libya}) \wedge \forall y ((y \neq \text{Libya}) \Rightarrow V(x, y))) \quad (\text{De Morgan's Law of Quantifiers})$$

$$\equiv \forall x (\neg (\neg V(x, \text{Libya}) \wedge \forall y ((y \neq \text{Libya}) \Rightarrow V(x, y)))) \quad (\text{1st De Morgan's Law})$$

$$\equiv \forall x (\neg (\neg V(x, \text{Libya})) \vee \neg \forall y ((y \neq \text{Libya}) \Rightarrow V(x, y)))$$

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$$\equiv \forall x (\neg (\neg V(x, \text{Libya})) \vee \neg \forall y ((y \neq \text{Libya}) \Rightarrow V(x, y))) \quad (\text{Double negation Law})$$

$$\equiv \forall x (V(x, \text{Libya}) \vee \neg \forall y ((y \neq \text{Libya}) \Rightarrow V(x, y))) \quad (\text{De Morgan's Law of Quantifiers})$$

$$\equiv \forall x (V(x, \text{Libya}) \vee \exists y \neg ((y \neq \text{Libya}) \Rightarrow V(x, y))) \quad (\text{Conditional-disjunction equivalence})$$

$$\equiv \forall x (V(x, \text{Libya}) \vee \exists y (\neg (\neg (y \neq \text{Libya})) \wedge \neg V(x, y))) \quad (\text{2nd De Morgan's Law})$$

$$\equiv \forall x (V(x, \text{Libya}) \vee \exists y ((y = \text{Libya}) \wedge \neg V(x, y))) \quad (\text{Double-negation Law})$$